



FAZE CLAN

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[46025596 – Z Cassim]

[46664734 – U Bada]

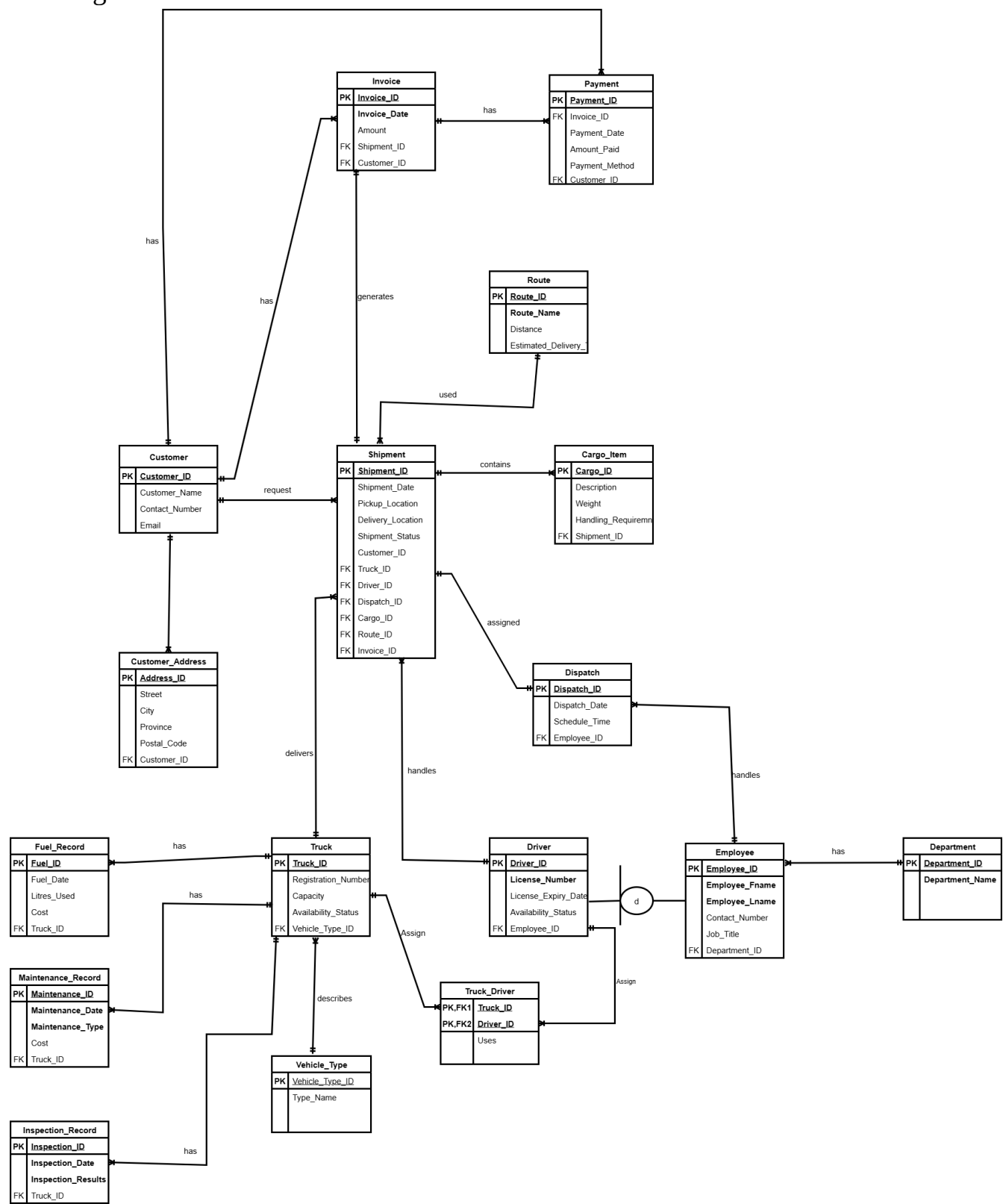
[53565509 – F Jhetam]

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Business Rules

1. Each customer may submit many shipment requests, but each shipment request belongs to one customer.
2. Each shipment must have one pickup location and one delivery location.
3. Each shipment may contain one or more cargo items, but each cargo item belongs to one shipment.
4. Each shipment is assigned to one dispatch record.
5. Each dispatch record is handled by one dispatch officer.
6. Each shipment is assigned one truck for delivery, but a truck can be assigned to many shipments over time.
7. Each shipment is assigned one driver, but a driver can handle many shipments over time.
8. Each shipment follows one route, but one route may be used for many shipments.
9. A driver may drive different trucks over time, and a truck may be driven by different drivers over time.
10. Each truck belongs to one vehicle type, but one vehicle type may describe many trucks.
11. Each truck may have many maintenance records, but each maintenance record belongs to one truck.
12. Each truck may have many inspection records, but each inspection record belongs to one truck.
13. Each truck may have many fuel usage records, but each fuel usage record belongs to one truck.
14. Each employee belongs to one department, but one department may have many employees.
15. Each completed shipment generates one invoice.
16. Each invoice belongs to one shipment and one customer.
17. One invoice may have one or many payment records until the full balance is settled.
18. A customer may have many invoices and many payments over time.
19. Each shipment has a status that changes over time, such as scheduled, collected, in transit, stored, or delivered.
20. A driver must have a valid licence before being assigned to a shipment.
21. A truck must pass inspection before it can be dispatched.
22. A route records distance and estimated delivery time for planning and reporting purposes.

ER Diagram



ER diagram Key Features

Weak Entity - CUSTOMER_ADDRESS is a weak entity — it cannot exist without a CUSTOMER. Its existence depends entirely on the Customer_ID foreign key.

Multi-valued Attribute - A Customer can have multiple addresses stored in CUSTOMER_ADDRESS, representing a multi-valued address attribute.

Mandatory Participation - Each SHIPMENT must have exactly one CUSTOMER, TRUCK, DRIVER, DISPATCH, and ROUTE (mandatory participation on the Shipment side).

Derived Attribute - Total outstanding balance on an INVOICE can be derived from summing PAYMENT.Amount_Paid records linked to that invoice.

Composite Attribute - Employee full name is composed of Employee_Fname and Employee_Lname, representing a composite name attribute.

Strong Relationship – MAINTENANCE_RECORD, INSPECTION_RECORD and FUEL_RECORD are dependent on TRUCK and cannot exist without the record

Composite Key – Truck_ID and Driver_ID are composite keys of TRUCK_DRIVER

Logical Design

INVOICE(Invoice_ID(PK), Invoice_Date, Amount, Shipment_ID(FK), Customer_ID(FK))

- Primary Key: Invoice_ID
- Foreign Key: Shipment_ID REFERENCES SHIPMENT
- Foreign Key: Customer_ID REFERENCES CUSTOMER

PAYMENT(Payment_ID(PK), Invoice_ID(FK), Payment_Date, Amount_Paid, Payment_Method, Customer_ID(FK))

- Primary Key: Payment_ID
- Foreign Key: Invoice_ID REFERENCES INVOICE
- Foreign Key: Customer_ID REFERENCES CUSTOMER

ROUTE(Route_ID(PK), Route_Name, Distance, Estimated_Delivery_Time)

- Primary Key: Route_ID

CUSTOMER(Customer_ID(PK), Customer_Name, Contact_Number, Email)

- Primary Key: Customer_ID

SHIPMENT(Shipment_ID(PK), Shipment_Date, Pickup_Location , Delivery_Location , Shipment_Status, Customer_ID(FK), Driver_ID(FK), Truck_ID(FK), Route_ID(FK), Dispatch_ID(FK), Invoice_ID(FK))

- Primary Key: Shipment_ID
- Foreign Key: Invoice_ID REFERENCES INVOICE
- Foreign Key: Customer_ID REFERENCES CUSTOMER
- Foreign Key: Driver_ID REFERENCES DRIVER
- Foreign Key: Truck_ID REFERENCES TRUCK
- Foreign Key: Route_ID REFERENCES ROUTE
- Foreign Key: Dispatch_ID REFERENCES DISPATCH

CARGO_ITEM(Cargo_ID(PK),Description, Weight, Handling_Requirements, Shipment_ID(FK))

- Primary Key: Cargo_ID
- Foreign Key: Shipment_ID REFERENCES SHIPMENT

CUSTOMER_ADDRESS(Address_ID(PK), Street, City, Province, Postal_Code, Customer_ID(FK))

- Primary Key: Address_ID
- Foreign Key: Customer_ID REFERENCES CUSTOMER

DISPATCH(Dispatch_ID(PK), Dispatch_Date, Schedule_Time, Employee_ID(FK))

- Primary Key: Dispatch_ID
- Foreign Key: Employee_ID REFERENCES EMPLOYEE

FUEL_RECORD(Fuel_ID(PK), Fuel_Date, Litres_Used, Cost, Truck_ID(FK))

- Primary Key: Fuel_ID
- Foreign Key: Truck_ID REFERENCES TRUCK

INSPECTION_RECORD(Inspection_ID(PK), Inspection_Date, Inspection_Result, Truck_ID(FK))

- Primary Key: Inspection_ID
- Foreign Key: Truck_ID REFERENCES TRUCK

MAINTENANCE_RECORD(Maintenance_ID(PK), Maintenance_Date, Maintenance_Type, Cost, Truck_ID(FK))

- Primary Key: Maintenance_ID
- Foreign Key: Truck_ID REFERENCES TRUCK

TRUCK(Truck_ID(PK), Registration_Number, Capacity, Availability_Status, VehicleType_ID(FK))

- Primary Key: Truck_ID
- Foreign Key: VehicleType_ID REFERENCES VEHICLE_TYPE

VEHICLE_TYPE(VehicleType_ID(PK), Type_Name)

- Primary Key: VehicleType_ID

DRIVER(Driver_ID(PK), Licence_Number, Licence_Expiry_Date, Availability_Status, Employee_ID(FK))

- Primary Key: Driver_ID
- Foreign Key: Employee_ID REFERENCES EMPLOYEE

TRUCK_DRIVER(Truck_ID(PKFK), DRIVER_ID(PKFK),Uses)

- Primary Key: Driver_ID
- Foreign Key: Driver_ID REFERENCES DRIVER
- Primary Key: Truck_ID
- Foreign Key: Truck_ID REFERENCES TRUCK

EMPLOYEE(Employee_ID(PK), First_Name, Last_Name, Contact_Number, Job_Title, Department_ID(FK))

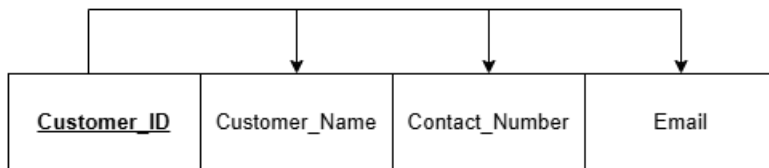
- Primary Key: Employee_ID
- Foreign Key: Department_ID REFERENCES DEPARTMENT

DEPARTMENT(Department_ID(PK), Department_Name)

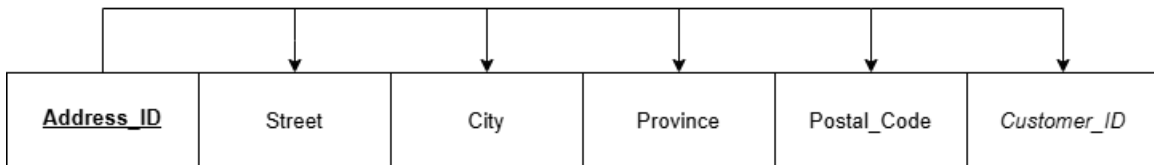
- Primary Key: Department_ID

Dependency Diagrams

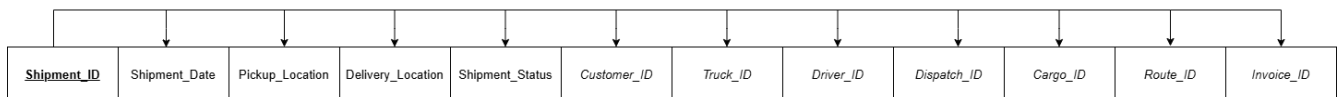
**Entity:
CUSTOMER**



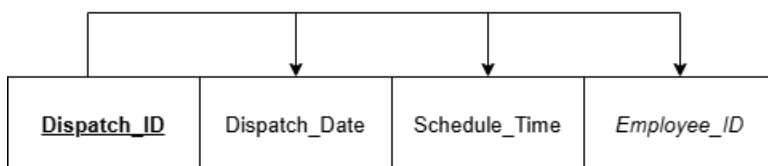
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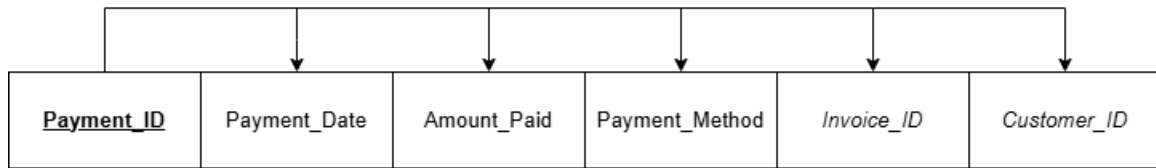
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SHIPMENT**



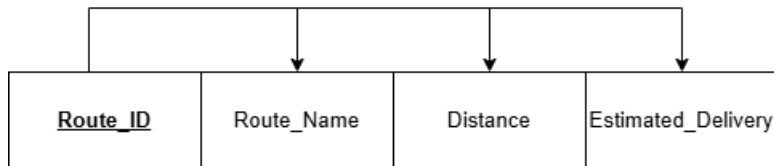
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DISPATCH**



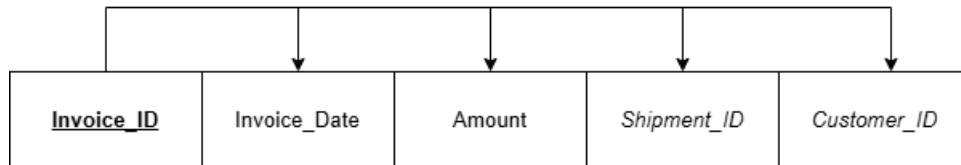
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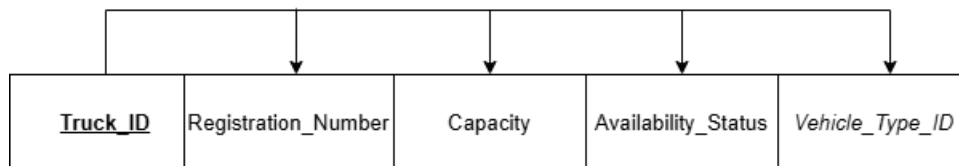
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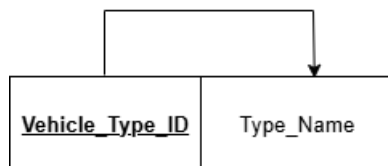
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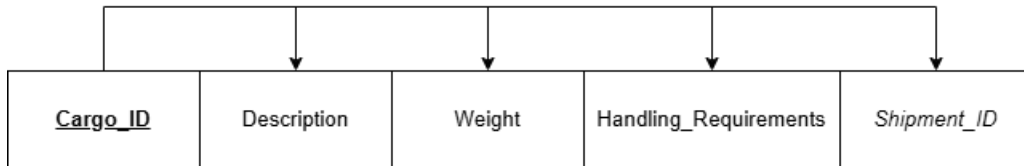
**Entity:
TRUCK**



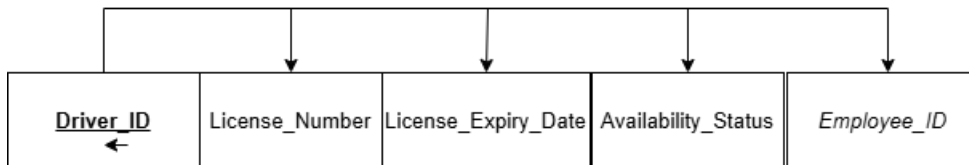
**Entity:
VEHICLE_TYPE**



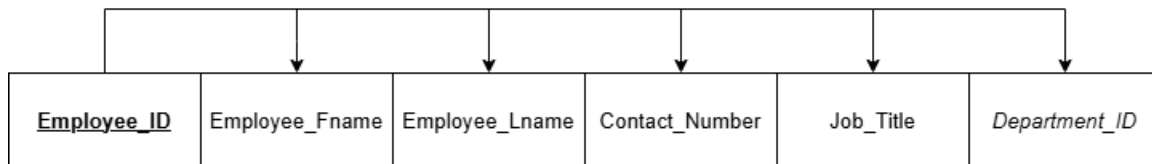
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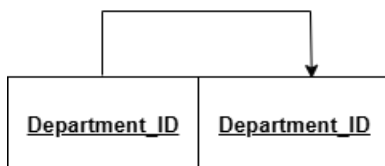
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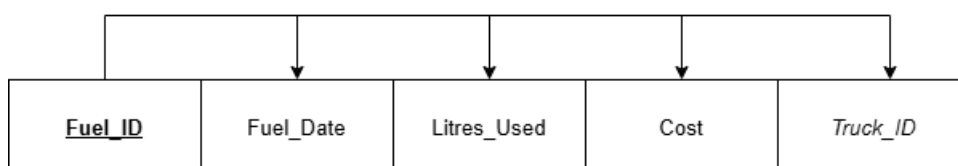
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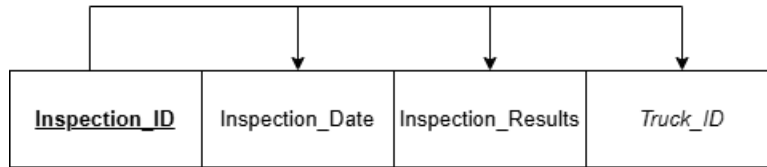
Entity:
DEPARTMENT



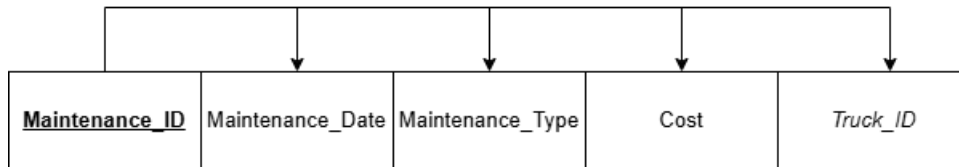
Entity:
FUEL_RECORD



Entity:
INSPECTION_RECORD



Entity:
MAINTENANCE_RECORD



Entity:
TRUCK_DRIVER

